

SECTION I (5 × 1 points)

Answer the questions below with a True or False.

1. The paths of the Wiener process are continuous.
2. The paths of the Wiener process are differentiable almost surely.
3. The multiperiod binomial model for stock prices is a martingale for all values of the coin toss probabilities.
4. A symmetric random walk almost surely returns to the origin given enough time.
5. A submartingale cannot decrease on average but has a tendency to increase on average.

SECTION II (10 points)

Relevant information: The random walk stochastic process is described by the sequence of random variables M_n which describe the location of the random walker at time n . The walker moves by ± 1 units at the i^{th} step if the i^{th} coin toss results in a heads/tails respectively. The underlying coin toss probabilities are p for heads and q for tails, with $p + q = 1$. The stock price process is described in terms of the random variable $S_n = u^{M_n} S_0$, where we have assumed that the up factor u and down factor d are related as $d = u^{-1}$.

1. (a) (4 points) By demanding that the *discounted* stock price process $Y_n = (1+r)^{-n} S_n$ is a martingale, derive the risk-neutral coin toss probabilities in terms of the up factor u , down factor $d = u^{-1}$ and the risk-free interest rate r . In other words, you must impose $\mathbb{E}_n[Y_{n+1}] = Y_n$. You can assume that there are only $n + 1$ coin tosses.
(b) (1 point) What must be the relationship between u , d and r for the risk-neutral probabilities to correspond to a fair coin?
2. (a) (3 points) It can be shown that the generating function for the probabilities $f_{2\nu}$ of first return to origin at the end of time 2ν for an asymmetric random walk is

$$F(\alpha) = \sum_{\nu=0}^{\infty} f_{2\nu} \alpha^{2\nu} = 1 - \sqrt{1 - 4pq\alpha^2}.$$

Show that $F(1) = 1 - |p - q|$. Interpret the value of $F(1)$ in terms of total probability of return to origin, when $p \neq q$ and also when $p = q = 1/2$.

- (b) (2 points) The derivative of the generating function evaluated at 1 gives the expectation value of the time that the walker returns to the origin. Compute this expectation value. Interpret the answer for $p = q = 1/2$.

$u^2 \frac{1}{u}$

SECTION I (5 × 1 points)

Answer the questions below with a True or False.

1. An Itô integral with respect to the Wiener process is always a martingale.
2. For the random walk process M_n , $0 \leq n \leq N$, the σ -algebra $\sigma(M_m)$ is a sub σ -algebra of $\sigma(M_n)$ for $m < n$.
3. If two probability measures on a probability space are equivalent, then they are related by a Radon-Nikodým derivative.
4. An agent hedging a short position on a European call priced according to the Black-Scholes-Merton solution must borrow (as opposed to invest) from the money market at each time instant.
5. The (non-discounted) stock price process given by geometric Brownian motion is a martingale.

SECTION II (10 points)

The Black-Scholes-Merton solution for the price of a European put at strike price K and expiration time T , based on a stock with mean rate of return α and volatility σ is

$$p(t, x) = Ke^{-r(T-t)}N(-d_-(T-t, x)) - xN(-d_+(T-t, x)),$$

where r is the risk-free interest rate in the money market, and

$$N(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-y^2/2} dy, \quad \text{and} \quad d_{\pm}(\tau, x) = \frac{1}{\sigma\sqrt{\tau}} \left(\log \frac{x}{Ke^{-r\tau}} \right) \pm \frac{\sigma\sqrt{\tau}}{2}.$$

6. (3 + 3 points) Compute the function $p(t, x)$ in the limit where the volatility of the stock is (a) zero ($\sigma \rightarrow 0$), and (b) infinite ($\sigma \rightarrow +\infty$). Interpret your results in one or two sentences.
7. (2 + 2 points) Let $\mathbb{X}$ be a discrete random variable which never takes the value zero and has mean 0 and variance 1.
 - a) Show that the random variable $\mathbb{Z} = \mathbb{X}^2$ satisfies the two properties required to qualify as a Radon-Nikodým derivative. If the maximum and minimum values of $\mathbb{X}$ are ± 2 respectively, and they occur with positive probability, can $\mathbb{X} + 1$ qualify as a Radon-Nikodým derivative?
 - b) Compute the mean and variance of the random variable $1/\mathbb{X}$ with respect to the new probability measure $\tilde{\mathbb{P}}$ induced by the Radon-Nikodým derivative $\mathbb{Z} = \mathbb{X}^2$.

Instructions

1. There are two sections: Section I on Math Background, and Section II on Applications to Derivatives Pricing.
 2. The exam is for 40 points with each section worth 20 points.
 3. Each section has 3 questions worth 10 points each. All questions are **compulsory**. The best 2 out of 3 answers in each section will be taken for your final score.
 4. This is a closed book exam. No notes/textbooks or other reference material, digital or otherwise, are allowed in the exam hall.
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SECTION I: MATH BACKGROUND

1. (5×2 points) Consider a random variable X whose range is $\{-2, -1, 1, 2\}$ with the probabilities

$$\mathbb{P}\{X = -2\} = \frac{1}{4}, \quad \mathbb{P}\{X = -1\} = \frac{3}{8}, \quad \mathbb{P}\{X = 1\} = \frac{1}{8}, \quad \mathbb{P}\{X = 2\} = \frac{1}{4}. \quad (1)$$

- a) Verify that $\mathbb{P}$ is a good probability measure. Which is the most probable value of X with respect to $\mathbb{P}$?
 - b) Compute $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$.
 - c) Let $Z = \alpha X^2$. For what value of α is Z a Radon-Nikodým derivative?
 - d) Let $\tilde{\mathbb{P}}$ be the new probability measure induced by the Radon-Nikodým derivative Z . What are the probabilities under $\tilde{\mathbb{P}}$ for X to take the different values in its range?
 - e) Which is the most probable value of X with respect to $\tilde{\mathbb{P}}$?
2. ($2 + 4 + 2 + 2$ points) In the questions below, $W(t)$ stands for Brownian motion (also known as the Wiener process).
- a) It is known that $W(t)^4$ can be written as the sum of a Itô integral and an ordinary integral:

$$W(t)^4 = \int_0^t f(W(u)) dW(u) + \int_0^t g(W(u)) du, \quad (2)$$

for some functions $f(x)$ and $g(x)$. Use the Itô-Doebelin formula for $d(W(t)^4)$ and obtain the functions $f(x)$ and $g(x)$.

- b) Calculate $\mathbb{E}[W(t)^4]$ by computing the expectation value on both sides of the integral formula above, after determining the functions f and g . Your final answer should be an explicit formula without involving integrals or expectations.

Hint: You need to recall the variance of $W(t)$ from the definition of the Wiener process at some stage of the calculation, and use the property that the expectation value of an ordinary integral is the integral of the expectation value.

- c) Calculate the variance $\text{Var}[W(t)^2]$ using the answer of part (b).
 d) Consider two Wiener processes W_1 and W_2 which satisfy

$$dW_1 dW_1 = dt, \quad dW_2 dW_2 = dt, \quad dW_1 dW_2 = \rho dt, \quad (3)$$

where ρ is a constant between -1 and 1 .

Calculate the covariance $\mathbb{E}[W_1(t)W_2(t)]$ by writing down the Itô product formula for $d(W_1 W_2)$, integrating with respect to t and then taking expectation values.

3. (2 + 2 + 3 + 3 points) Consider the Wiener process $W(t)$ for the time period $0 \leq t \leq T$, and its associated filtration $\{\mathcal{F}(t)\}_{0 \leq t \leq T}$ of σ -algebras. Let $f(x)$ be an ordinary function of one variable.

- a) Which property of the Wiener process allows us to write the equation

$$\mathbb{E}[f(W(T)) \mid \mathcal{F}(t)] = g(t, W(t)), \quad (4)$$

for some ordinary function $g(t, x)$?

Using the property of iterated conditional expectations, show that $g(t, W(t))$ is a martingale (even without knowledge of the function g).

- b) Using the Itô-Doebelin formula for dg and the property that $g(t, W(t))$ is a martingale, derive the partial differential equation satisfied by $g(t, x)$.

- c) Consider $f(x) = e^x$. Compute $\mathbb{E}[f(W(T)) \mid \mathcal{F}(t)]$ explicitly by writing $W(T) = W(T) - W(t) + W(t)$ and using independence of non-overlapping increments and other standard properties of the Wiener process.

You may need the result $\mathbb{E}[e^X] = e^{\sigma^2/2}$ where X is a normal random variable with mean 0 and variance σ^2 .

- d) Read off the function $g(t, x)$ by replacing $W(t)$ by x in the previous answer for $\mathbb{E}[e^{W(T)} \mid \mathcal{F}(t)]$, and verify that this $g(t, x)$ satisfies the partial differential equation derived in part (b).

SECTION II: APPLICATIONS TO DERIVATIVES PRICING

1. (5 × 2 points) Let us focus on the Black-Scholes-Merton solution for the price of a European call with strike price K and expiration time T , based on a single stock with constant mean return α and volatility σ . The price at time t and stock value $x = S(t)$ is

$$c(t, x) = xN(d_+(T-t, x)) - Ke^{-r(T-t)}N(d_-(T-t, x)) , \quad (5)$$

where r is the risk-free interest rate in the money market, and

$$N(y) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^y e^{-z^2/2} dz , \quad \text{and} \quad d_{\pm}(\tau, x) = \frac{1}{\sigma\sqrt{\tau}} \left(\log \frac{x}{Ke^{-r\tau}} \right) \pm \frac{\sigma\sqrt{\tau}}{2} .$$

- a) The two terms in $c(t, x)$ can be interpreted as a simultaneous long position on $\Delta(t)$ shares of the stock $S(t)$ and a short position on $M(t)$ zero coupon bonds with face value K and maturity time T .

Read off the number of shares $\Delta(t)$ and number of zero coupon bonds $M(t)$ from the price function $c(t, x)$. Be careful of discounting factors when you do this.

- b) For any fixed time t , how do the functions $d_{\pm}(T-t, x)$ behave for really small ($x \rightarrow 0$) and really large ($x \rightarrow \infty$) stock prices?
- c) How do $\Delta(t)$ and $M(t)$ from part (a) behave for low and high stock prices?
- d) Repeat parts (a) and (c) for a European put on the same stock, at the same strike price K and expiration time T , with price function

$$p(t, x) = Ke^{-r(T-t)}N(-d_-(T-t, x)) - xN(-d_+(T-t, x)) . \quad (6)$$

- e) Repeat parts (a) and (c) for a long forward contract on the same stock, at delivery price K and delivery time T , with price function

$$f(t, x) = x - Ke^{-r(T-t)} . \quad (7)$$

2. (5 × 2 points) Let us take a market model with risk-free interest rate r , one Brownian motion $W(t)$, and two stocks $S(t)$ and $Q(t)$ governed by the stochastic differential equations

$$dS(t) = \alpha_1 S(t)dt + \sigma_1 S(t)dW(t) , \quad dQ(t) = \alpha_2 Q(t)dt + \sigma_2 Q(t)dW(t) . \quad (8)$$

- a) Obtain the differential of $1/Q(t)$ by applying the Itô-Doebelin formula to $f(Q(t))$ with $f(x) = 1/x$.
- b) Consider the shifted Brownian motion $\widetilde{W}(t) = W(t) + \theta t$ where θ is a constant. Write down the *market price of risk equations* that arise from demanding that the discounted stocks $e^{-rt}S(t)$ and $e^{-rt}Q(t)$ are martingales with respect to the new probability measure under which $\widetilde{W}$ is standard Brownian motion.

- c) Find out the algebraic relation between $\alpha_1, \alpha_2, \sigma_1, \sigma_2, r$ for the market price of risk equations to have a solution for θ .

When is arbitrage possible and when is it not possible in this model? Justify your answer.

- d) According to Girsanov's theorem, the θ you found in part (b) allows us to transform to the risk-neutral measure with respect to which $\widetilde{W}(t)$ is standard Brownian motion. The stock prices now satisfy

$$dS(t) = rS(t)dt + \sigma_1 S(t)d\widetilde{W}(t), \quad dQ(t) = rQ(t)dt + \sigma_2 Q(t)d\widetilde{W}(t). \quad (9)$$

Consider the relative stock price $\frac{S(t)}{Q(t)}$. By using the above differentials and the Itô product rule for the product $S(t)\frac{1}{Q(t)}$, obtain the stochastic differential equation for the relative stock price.

- e) Let $\widehat{W}(t) = \widetilde{W}(t) + \eta t$ where η is a constant. Find the value of η which makes the relative stock price process a martingale with respect to the new measure under which $\widehat{W}(t)$ is a standard Brownian motion.

3. (2 + 2 + 3 + 3 points) Consider a market model with one stock S_n and the risk-free interest rate process R_n governed by the following N -period binomial tree models (with the n^{th} coin toss denoted by ω_n):

$$S_{n+1} = \begin{cases} uS_n & \text{if } \omega_n = H \\ dS_n & \text{if } \omega_n = T \end{cases} \quad \text{and} \quad R_n = \begin{cases} r_1 & \text{if } \omega_n = H \\ r_2 & \text{if } \omega_n = T \end{cases}, \quad (10)$$

for $n = 1, 2, \dots, N-1$. The initial (time 0) stock price is denoted S_0 and the initial interest rate is denoted R_0 . These are not random variables since they do not depend on the outcome of any coin toss.

Note that R_n determines the interest rate between time n and $n+1$. So, the interest rate between time 0 and 1 is R_0 , between time 1 and 2 is R_1 and so on.

$\alpha = R_0 = \sum$

Let us work with a 3-period binomial tree ($N = 3$) with the following parameters:

$$S_0 = 4, \quad u = d^{-1} = 2, \quad R_0 = 1/4, \quad r_1 = 1/2, \quad r_2 = 1/4. \quad (11)$$

- Draw the 3-period binomial tree diagram showing the stock price and interest rate at each node.
- Calculate the risk-neutral probabilities for each branch of the binomial tree and add them to the tree diagram in part (a). (You can draw a separate diagram if you wish.)

Hint: In this model, the coin tosses are not independent of each other and the probabilities are path dependent. Hence, there are separate risk-neutral probabilities for each branch in the binomial tree. The risk-neutral probabilities for the H and T branches coming out of a particular node are given by the usual formula

$$\tilde{p} = \frac{1 + r - d}{u - d}, \quad \tilde{q} = \frac{u - (1 + r)}{u - d}, \quad (12)$$

where r is now the risk-free interest rate attached to that particular node.

- Using the stock prices and risk neutral probabilities computed in parts (a) and (b), compute the risk-neutral price at time 0 of a European call struck at $K = 5$ with expiry at time 2.

You can either use the risk-neutral pricing formula $V_0 = \tilde{\mathbb{E}}[\mathbb{D}_2(S_2 - K)^+]$ directly where $\mathbb{D}_n$ is the discount process $\mathbb{D}_n = (1 + R_0)^{-1} \cdots (1 + R_{n-1})^{-1}$, or start with the payoffs at time 2 and roll back period by period by computing the discounted average at each node as we did in class.

- The m -forward interest rate $F_{n,m}$ at time $n \leq m$ is $F_{n,m} = \frac{\mathbb{B}_{n,m} - \mathbb{B}_{n,m+1}}{\mathbb{B}_{n,m+1}}$ where $\mathbb{B}_{n,m}$ is the risk-neutral price of the bond with face value 1 and maturity time m . Compute $F_{1,2}$ (the 2-forward interest rate at time 1). How does this compare with the interest rate R_2 at time 2?

OR

Compute the risk-neutral price at time 0 of a time-2 interest rate caplet with fixed rate $K = 1/8$. (The time-2 caplet pays off $(R_1 - K)^+$ at time 2 and nothing at other times).